Triangle Inequality Theorem; Converse of the Pythagorean Theorem

Question Bank - Additional 6.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 3, can the three lengths make a triangle?

1. 4 m, 10 m, 12 m

Yes — 4 + 10 = 14 > 12

2. 6 inches, 17 inches, 9 inches

No — 6 + 9 = 15 < 17

The longest is listed second.

3. 14 ft, 7 ft, 7 ft

No — 7 + 7 = 14 lies flat

The equal case.

For questions 4 and 5, two sides of a triangle are given. Write an inequality for the third side, x.

4. 6 cm and 9 cm

3 < x < 15 (centimeters)

5. 1 inch and 8 inches; the third side is a whole number of inches — which one?

7 < x < 9, so x = 8 in — the only whole number between

For questions 6 through 8, do the lengths make a right triangle?

6. 2.4 m, 3.2 m, 4 m

Yes — 5.76 + 10.24 = 16 = 4^2

7. 39 inches, 36 inches, 15 inches

Yes — $15^2 + 36^2 = 1{,}521 = 39^2$

The longest is listed first.

8. 8 m, 9 m, 12 m

No — 64 + 81 = 145, not 144

Choose one.

9. Which set of lengths makes a right triangle?

- ○ A. 5, 6, 7
- ● **B. 14, 48, 50**
- ○ C. 9, 12, 16
- ○ D. 3, 4, 7

B

196 + 2,304 = 2,500 = 50^2.